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Marks 31.92/40.00

Grade 79.79 out of 100.00

Question 1

Complete

Not graded

By selecting "I confirm", I hereby declare under oath that I will work on this examination on my own without any help or any third-party assistance.

By selecting "I confirm", I understand that noncompliance results in invalidation of the assessment, whereby the invalidated examination will be added to the total number of retakes and noncompliance may result in further legal action.

- ☐ a. I do not confirm
- ☒ b. I confirm

The correct answer is: I confirm

Question 2

Partially correct

Mark 0.33 out of 1.00

Consider the following `Dataset` implementation:

```
import numpy as np
from torch.utils.data import Dataset

class MyDataset(Dataset):
    def __init__(self, seed):
        self.rng = np.random.default_rng(seed)

    def __getitem__(self, index):
        return self.rng.uniform(size=(5,))
```

Which of the following statements are true?

- ☐ a. An instance of this class would produce infinitely many samples.
- ☐ b. In a multi-processed data loading setting, the returned samples might not be reproducible because the seed of the random number generator is set only in the `__init__` method.
- ☐ c. The implementation is incorrect since the `__getitem__` method should return a PyTorch tensor and not a NumPy array.
- ☒ d. The `__len__` method should be implemented if it is intended to be used in combination with a default `DataLoader` (with default `Sampler`). ✓

The correct answers are: An instance of this class would produce infinitely many samples., In a multi-processed data loading setting, the returned samples might not be reproducible because the seed of the random number generator is set only in the `__init__` method., The `__len__` method should be implemented if it is intended to be used in combination with a default `DataLoader` (with default `Sampler`).

Question 3

Correct

Mark 1.00 out of 1.00

Which of the following are benefits of a version control system (VCS) such as git?

- ☒ a. Efficient tracking of files and their changes. ✓
- ☐ b. Automatic integration of unit tests.
- ☒ c. Access to previous versions. ✓
- ☒ d. Useful for collaboration. ✓

The correct answers are: Efficient tracking of files and their changes., Useful for collaboration., Access to previous versions.

Question 4

Incorrect

Mark 0.00 out of 1.00

Assume you want to normalize your data and compute a normalization constant from your dataset (e.g., the mean over many samples). What would be the correct way to proceed in terms of training, validation and test set?

- ☐ a. Determine the constant for normalization on the whole dataset and apply it to training, validation and test data.
- ☒ b. Determine the constant for normalization on the training and validation data and apply it also to the test data. ✖
- ☐ c. Determine the constant for normalization on the validation and test data and apply it also to the training data.
- ☐ d. Determine the constant for normalization on the training data and apply it also to the validation and test data.

The correct answer is: Determine the constant for normalization on the training data and apply it also to the validation and test data.

Question 5

Correct

Mark 1.00 out of 1.00

Assume you have the following binary confusion matrix:

		Predicted	
		positive	negative
Actual	positive	30	0
	negative	5	10

Which of the following statements are correct?

- ☒ a. The total number of samples is 45. ✔
- ☐ b. The used model is more likely to predict the negative class.
- ☐ c. The classification performance is perfect.
- ☒ d. There are 30 true positives. ✔

The correct answers are: There are 30 true positives., The total number of samples is 45.

Question 6

Correct

Mark 1.00 out of 1.00

When training a model using some gradient-based iterative method, how should the loss (computed with some loss function) change?

- ☐ a. The loss should nearly stay constant (given some user-defined value).
- ☐ b. The loss should be maximized.
- ☒ c. The loss should be minimized. ✓
- ☐ d. The loss is irrelevant when training a model

The correct answer is: The loss should be minimized.

Question 7

Correct

Mark 1.00 out of 1.00

Assume you have the following categorical data in exactly this fixed order:

```
"Spider"  
"Frog"  
"Fly"  
"Snail"
```

You are asked to create a one-hot encoding. Which of the following is the correct one-hot encoded representation of "Frog"?

- ☐ a. (1, 0, 0, 0)
- ☐ b. (0, 0, 0, 0)
- ☒ c. (0, 1, 0, 0) ✓
- ☐ d. (0, 0, 1, 0)
- ☐ e. (1, 1, 1, 1)
- ☐ f. (1, 0, 1, 1)
- ☐ g. (0, 0, 0, 1)

The correct answer is: (0, 1, 0, 0)

Question 8

Correct

Mark 1.00 out of 1.00

Assume you apply the following PyTorch image transformation to some example RGB image:

```
transforms.Compose([
    transforms.Grayscale(),
    transforms.RandomHorizontalFlip(0.5),
    transforms.RandomVerticalFlip(0.5),
    transforms.ToTensor()
])
```

Which of the following statements are correct?

- ☒ a. The resulting image might be both horizontally and vertically flipped. ✓
- ☒ b. The resulting image might not be flipped at all (neither horizontally nor vertically). ✓
- ☒ c. The resulting image will be a PyTorch tensor. ✓
- ☒ d. The resulting image might be vertically flipped. ✓
- ☒ e. The resulting image will be in grayscale. ✓
- ☒ f. The resulting image might be horizontally flipped. ✓

The correct answers are: The resulting image will be a PyTorch tensor., The resulting image will be in grayscale., The resulting image might be horizontally flipped., The resulting image might be vertically flipped., The resulting image might be both horizontally and vertically flipped., The resulting image might not be flipped at all (neither horizontally nor vertically).

Question 9

Correct

Mark 1.00 out of 1.00

What is the primary task of a hash function?

- ☒ a. To compute a fixed-sized data vector for a given input. ✓
- ☐ b. To compare image data.
- ☐ c. To compute whether two objects are equal.
- ☐ d. To make neural networks work for arbitrary input data.

The correct answer is: To compute a fixed-sized data vector for a given input.

Question 10

Correct

Mark 1.00 out of 1.00

Which of the following is the correct order of performing one weight update in PyTorch (you can assume that `optimizer` is a valid PyTorch optimizer object)?

- ☐ a. 1. Compute `loss`
2. Update weights `optimizer.step()`
3. Compute gradients `loss.backward()`
4. Reset gradients `optimizer.zero_grad()`
- ☒ b. 1. Compute `loss` ✓
2. Compute gradients `loss.backward()`
3. Update weights `optimizer.step()`
4. Reset gradients `optimizer.zero_grad()`
- ☐ c. 1. Reset gradients `optimizer.zero_grad()`
2. Compute `loss`
3. Update weights `optimizer.step()`
4. Compute gradients `loss.backward()`
- ☐ d. 1. Compute `loss`
2. Compute gradients `loss.backward()`
3. Reset gradients `optimizer.zero_grad()`
4. Update weights `optimizer.step()`

The correct answer is: 1. Compute `loss`
2. Compute gradients `loss.backward()`
3. Update weights `optimizer.step()`
4. Reset gradients `optimizer.zero_grad()`

Question 11

Correct

Mark 1.00 out of 1.00

Which of the following statements are true regarding early stopping?

- ☒ a. The stopping can be achieved by some heuristic (e.g., no validation loss improvement for a defined number of updates/epochs). ✓
- ☒ b. Using some validation set, early stopping can be applied to cancel/stop the training early. ✓
- ☐ c. Early stopping is done automatically if the training loss reaches 0.
- ☐ d. Early stopping guarantees to return the best possible model, i.e., no better model could have been found if the training had continued.

The correct answers are: Using some validation set, early stopping can be applied to cancel/stop the training early., The stopping can be achieved by some heuristic (e.g., no validation loss improvement for a defined number of updates/epochs).

Question 12

Partially correct

Mark 0.67 out of 1.00

Which of the following statements are true regarding minibatch learning?

- ☒ a. The batch size is a hyperparameter. ✓
- ☐ b. Minibatch learning can potentially benefit from broadcast computations.
- ☐ c. The bigger the batch size, the better the learning (in terms of predictive model performance).
- ☒ d. The samples within a minibatch are used for one update step. ✓

The correct answers are: The batch size is a hyperparameter., The samples within a minibatch are used for one update step., Minibatch learning can potentially benefit from broadcast computations.

Question 13

Partially correct

Mark 0.67 out of 1.00

Assume a classification task where you want to predict the whether it is day or night based on (nature) images. Which of the following data augmentation techniques might be problematic?

- ☒ a. flipping horizontally ✗
- ☒ b. color jittering ✓
- ☐ c. rotating by +-5 degrees
- ☐ d. adding slight noise

The correct answer is: color jittering

Question 14

Partially correct

Mark 0.50 out of 1.00

When creating minibatches and trying to stack input arrays in PyTorch (using `torch.utils.data.DataLoader`), it can happen that not all arrays of a batch are of the same shape. Which of the following statements are correct?

- ☐ a. Stacking can be skipped and the arrays can just be kept in a list.
- ☐ b. A `DataLoader` cannot actually be used in this case.
- ☐ c. Arrays of unequal shape must be dropped/excluded.
- ☒ d. Padding can be applied to equalize all array shapes. ✓

The correct answers are: Padding can be applied to equalize all array shapes., Stacking can be skipped and the arrays can just be kept in a list.

Question 15

Correct

Mark 1.00 out of 1.00

Assume that `MyModule` is a class properly derived from the `torch.nn.Module` class and `tensor_a` is a PyTorch tensor. What does the following code do?

```
my_module = MyModule()  
my_module(tensor_a)
```

- ☐ a. It creates an instance of `MyModule`. Then it (ultimately) calls the `__getitem__` method of `my_module` with argument `tensor_a`.
- ☐ b. It creates an instance of `MyModule`. Then it reads a sample into `tensor_a`.
- ☐ c. It raises an exception, since one cannot create instances of `MyModule`.
- ☒ d. It creates an instance of `MyModule`. Then it (ultimately) calls the `forward` method of `my_module` with argument `tensor_a`. ✓

The correct answer is: It creates an instance of `MyModule`. Then it (ultimately) calls the `forward` method of `my_module` with argument `tensor_a`.

Question 16

Correct

Mark 1.00 out of 1.00

Consider the following incomplete implementation of a custom `Dataset` which should return samples from an already existing numpy array `samples` and (in addition) an ID for debugging purposes:

```
import numpy as np
from torch.utils.data import Dataset

# numpy array of shape (N, F), where
# N = number of samples and
# F = number of features per sample
samples = ...

class MyDataset(Dataset):
```

Which of the following is the correct implementation of the missing `__getitem__` method? For simplicity, the sample ID is just based on the provided `index`.

- ☐ a.

```
def __getitem__(self, index):
    return np.array([samples[index], index])
```
- ☐ b.

```
def __getitem__(self, index):
    return np.array([samples, index])
```
- ☐ c.

```
def __getitem__(self, index):
    return samples, index
```
- ☒ d.

```
def __getitem__(self, index):
    return samples[index], index
```

 ✓

The correct answer is:

```
def __getitem__(self, index):
    return samples[index], index
```

Question 17

Correct

Mark 1.00 out of 1.00

Why is the following code problematic (you can assume correct inputs, shapes and arguments)?

```
import torch

class MyModule(torch.nn.Module):
    def __init__(self):
        super().__init__()
        self.linear_list = [
            torch.nn.Linear(...),
            torch.nn.Sigmoid(),
            torch.nn.Linear(...)
        ]

    def forward(self, x):
        out = x
        for module in self.linear_list:
            out = module(out)
        return out
```

- ☐ a. `linear_list` is not applied like a `torch.nn.Module` should, i.e., the code in the `forward` method should be `out = linear_list(x)`.
- ☒ b. The attribute `linear_list` is a plain Python list and will thus be ignored by the automatic parameter registration. ✓
- ☐ c. `linear_list` should be created in the `forward` method instead of the `__init__` method.
- ☐ d. Just like `torch.nn.Sigmoid`, `torch.nn.Linear` must only be applied once.

The correct answer is: The attribute `linear_list` is a plain Python list and will thus be ignored by the automatic parameter registration.

Question 18

Partially correct

Mark 0.33 out of 1.00

Which statements regarding PyTorch's parameter registration of a `torch.nn.Module` are true?

- ☒ a. When assigned as attribute, a `torch.nn.Parameter` will automatically be registered. ✓
- ☐ b. A `torch.nn.Module` must always contain at least one (trainable) parameter.
- ☐ c. When assigned as attribute, a `torch.nn.Module` (or submodule) will automatically be registered.
- ☐ d. Parameters and modules can be registered manually.

The correct answers are: When assigned as attribute, a `torch.nn.Parameter` will automatically be registered., When assigned as attribute, a `torch.nn.Module` (or submodule) will automatically be registered., Parameters and modules can be registered manually.

Question 19

Partially correct

Mark 0.67 out of 1.00

In PyTorch, why can the gradient computation and the creation of the computational graph typically be disabled in the evaluation phase?

- ☐ a. The gradients during evaluation are approximately zero, so they can be replaced with the constant 0 to reduce computational run time.
- ☐ b. The gradients are the same during evaluation, so they can be copied from the training phase.
- ☒ c. Computing the gradients during evaluation would change the model parameters. ✖
- ☒ d. The model parameters are not updated during evaluation, so computing gradients is unnecessary. ✔

The correct answer is: The model parameters are not updated during evaluation, so computing gradients is unnecessary.

Question 20

Correct

Mark 1.00 out of 1.00

In a supervised setting, what does the term "machine learning model" mean?

- ☐ a. It represents the procedure to select different hyperparameter values.
- ☐ b. It represents the process of training and evaluating an algorithm.
- ☒ c. It represents a parameterized function/method that produces a (model) output given some input data. ✔
- ☐ d. It represents a gradient-descent based algorithm.

The correct answer is: It represents a parameterized function/method that produces a (model) output given some input data.

Question 21

Incorrect

Mark 0.00 out of 1.00

What is the output of the following code?

```
import torch
from torch.utils.data import Dataset, DataLoader

class MyDataset(Dataset):
    def __getitem__(self, index):
        return torch.tensor([10, 20])

    def __len__(self):
        return 6

dataset = MyDataset()
loader = DataLoader(dataset, batch_size=3)

for i, x in enumerate(loader):
    print(f"{i}: {tuple(x.shape)}")
```

- ☐ a. 0: (3, 3, 2)
- ☐ b. 0: (6, 2)
1: (6, 2)
- ☐ c. 0: (3, 2)
1: (3, 2)
- ☐ d. 0: (3, 6, 2)
- ☐ e. 0: (2,)
1: (2,)
2: (2,)
- ☒ f. 0: (6, 2) ✖
1: (6, 2)
2: (6, 2)
- ☐ g. 0: (3, 2)
1: (3, 2)
2: (3, 2)
3: (3, 2)
4: (3, 2)
5: (3, 2)

The correct answer is:

```
0: (3, 2)
1: (3, 2)
```

Question 22

Correct

Mark 1.00 out of 1.00

When training a model using some gradient-based iterative method, what does it mean when the loss (computed with some loss function) approaches zero?

- ☐ a. It means that the model is not learning.
- ☐ b. It does not mean anything, since the loss is irrelevant when training a model.
- ☐ c. It means that the model is overfitting.
- ☒ d. It means that the model is able to compute outputs that are close to the true target. ✓

The correct answer is: It means that the model is able to compute outputs that are close to the true target.

Question 23

Correct

Mark 1.00 out of 1.00

Consider a binary classification task where you want to classify sensors into either working (negative class) or faulty (positive class). There are roughly equally many working as faulty sensors. Which of the following evaluation metrics are appropriate for measuring the model performance?

- ☒ a. accuracy (ACC) ✓
- ☐ b. mean-squared error (MSE)
- ☐ c. mean absolute error (MAE)
- ☒ d. true positive rate (TPR) ✓

The correct answers are: accuracy (ACC), true positive rate (TPR)

Question 24

Correct

Mark 1.00 out of 1.00

Which of the following are commonly used loss functions?

- ☐ a. rectified linear unit (ReLU)
- ☐ b. sigmoid
- ☒ c. mean-squared error ✓
- ☒ d. cross entropy ✓

The correct answers are: cross entropy, mean-squared error

Question 25

Incorrect

Mark 0.00 out of 1.00

Which of the following statements are correct regarding the following hash function:

```
def my_hash(x):  
    return 304
```

- ☐ a. The hash value is the same in different Python interpreter sessions.
- ☒ b. Up to 304 hash collisions can occur for unequal inputs. ✖
- ☐ c. The hash function will work on any input.
- ☒ d. The computation of the hash value is fast. ✔

The correct answers are: The computation of the hash value is fast., The hash value is the same in different Python interpreter sessions., The hash function will work on any input.

Question 26

Partially correct

Mark 0.67 out of 1.00

When creating data splits such as training, validation and test sets, what should be taken into account?

- ☒ a. The splits should be reproducible. ✔
- ☐ b. The splits should be equally sized.
- ☒ c. The splits should be normalized. ✖
- ☐ d. The splits should not be equally sized.

The correct answer is: The splits should be reproducible.

Question 27

Incorrect

Mark 0.00 out of 1.00

What is the output of the following code?

```
import torch

def function(x: torch.Tensor):
    return x if x.max() > 5 else x * -1

scripted_function = torch.jit.script(function)
traced_function = torch.jit.trace(function, example_inputs=torch.tensor([4, 5, 6]))

actual_input = torch.tensor([1, 2, 3])
print("s:", scripted_function(actual_input).tolist())
print("t:", traced_function(actual_input).tolist())
```

- ☐ a. s: [-1, -2, -3]
t: [-4, -5, -6]
- ☐ b. s: [1, 2, 3]
t: [-1, -2, -3]
- ☐ c. s: [1, 2, 3]
t: [4, 5, 6]
- ☐ d. s: [-1, -2, -3]
t: [1, 2, 3]
- ☐ e. s: [1, 2, 3]
t: [-4, -5, -6]
- ☒ f. s: [-1, -2, -3] ✖
t: [4, 5, 6]
- ☐ g. s: [1, 2, 3]
t: [1, 2, 3]
- ☐ h. s: [-1, -2, -3]
t: [-1, -2, -3]

The correct answer is:

```
s: [-1, -2, -3]
t: [1, 2, 3]
```

Question 28

Correct

Mark 1.00 out of 1.00

In a survey, you collected people's favorite English words (some examples: "house", "tiger", "relaxing", etc.). Which type of data is this?

- ☐ a. ordinal data
- ☐ b. continuous data
- ☐ c. discrete data
- ☒ d. categorical data ✓

The correct answer is: categorical data

Question 29

Correct

Mark 1.00 out of 1.00

Which of the following statements are correct regarding a class that derives from `torch.nn.Module`?

- ☐ a. The class must not contain other `Modules`.
- ☐ b. The `__getitem__` method returns the training samples.
- ☒ c. The `forward` method specifies how an input is transformed into an output (the "flow" through the architecture). ✓
- ☒ d. The `__init__` method sets up the model architecture. ✓

The correct answers are: The `forward` method specifies how an input is transformed into an output (the "flow" through the architecture), The `__init__` method sets up the model architecture.

Question 30

Correct

Mark 1.00 out of 1.00

Which of the following are typical data normalization/scaling approaches?

- ☐ a. Scaling to range [1, 2].
- ☒ b. Scaling to range [-1, 1]. ✓
- ☒ c. Scaling to range [0, 1]. ✓
- ☒ d. Scaling to zero (0) mean and unit (1) variance. ✓

The correct answers are: Scaling to range [0, 1], Scaling to range [-1, 1], Scaling to zero (0) mean and unit (1) variance.

Question 31

Correct

Mark 1.00 out of 1.00

What is the output of the following code?

```
import torch
import torch.nn as nn

class MyModule(nn.Module):
    def __init__(self):
        super().__init__()
        self.linear = nn.Linear(in_features=2, out_features=3)

    def forward(self, x):
        out = self.linear(x)
        return torch.sigmoid(out)

inp = torch.rand(size=(16, 2))
my_module = MyModule()
result = my_module(inp)
print(tuple(result.shape))
```

- ☐ a. (16, 2)
- ☐ b. (16, 3, 2)
- ☐ c. (16,)
- ☐ d. (32,)
- ☐ e. (48,)
- ☒ f. (16, 3) ✓
- ☐ g. (16, 2, 3)

The correct answer is:

(16, 3)

Question 32

Partially correct

Mark 0.67 out of 1.00

Which of the following statements are correct regarding monitoring a model during training?

- ☐ a. Monitoring might decrease the model performance.
- ☒ b. Monitoring might aid in determining over- or underfitting. ✓
- ☐ c. Monitoring might lead to higher computational run times.
- ☒ d. Monitoring might help in finding issues during training. ✓

The correct answers are: Monitoring might lead to higher computational run times., Monitoring might help in finding issues during training., Monitoring might aid in determining over- or underfitting.

Question 33

Correct

Mark 1.00 out of 1.00

The special method `__getitem__` in a `torch.utils.data.Dataset` derived class should...

- ☐ a. ...return all samples.
- ☐ b. ...return the index of a specified sample.
- ☒ c. ...return one sample. ✓
- ☐ d. ...return minibatched samples.

The correct answer is: ...return one sample.

Question 34

Correct

Mark 1.00 out of 1.00

Numerical data can be described as:

- ☒ a. Quantitative data with mathematical meaning and a natural ordering. ✓
- ☐ b. Quantitative data with mathematical meaning but without a natural ordering.
- ☐ c. Qualitative data without mathematical meaning but with a natural ordering.
- ☐ d. Qualitative data without mathematical meaning.

The correct answer is: Quantitative data with mathematical meaning and a natural ordering.

Question 35

Correct

Mark 1.00 out of 1.00

What is the output of the following code?

```
import torch

class MyModule(torch.nn.Module):
    def __init__(self, a):
        super().__init__()
        self.a = a

    def forward(self, x):
        if x > 0:
            output = x + self.a
        else:
            output = x - self.a
        return output

my_module = MyModule(2.0)
c = my_module(5.0)
print(c)
```

- ☒ a. 7.0 ✓
- ☐ b. It raises an exception because control flow is not allowed in the `forward` method.
- ☐ c. It raises an exception at `self.a = a` because only PyTorch trainable parameters and (sub)modules can be used as attributes.
- ☐ d. It raises an exception because the `forward` method is not called.

The correct answer is: 7.0

Question 36

Correct

Mark 1.00 out of 1.00

Assume you have a training set and a test set. Which of the following statements are correct regarding data augmentation and these two sets?

- ☒ a. The test set should not be augmented since only real data should be used for evaluation. ✓
- ☐ b. The bigger the training set is compared to the test set, the fewer augmentations are needed.
- ☐ c. The training set should not be augmented since only real data should be used for training.
- ☐ d. The data augmentations must be the same for both the training set and the test set.

The correct answer is: The test set should not be augmented since only real data should be used for evaluation.

Question 37

Partially correct

Mark 0.67 out of 1.00

Training a neural network for one update will...

- ☐ a. ...perform training until the model overfits on all training samples.
- ☒ b. ...perform one weight update. ✓
- ☐ c. ...perform training for a fixed number of seconds.
- ☒ d. ...perform one training iteration over all training samples. ✗

The correct answer is: ...perform one weight update.

Question 38

Correct

Mark 1.00 out of 1.00

Which of the following data augmentation techniques can be applied to image data?

- ☒ a. dropping out pixels ✓
- ☒ b. rotating ✓
- ☒ c. adding noise ✓
- ☒ d. zooming and/or cropping ✓
- ☒ e. color jittering ✓
- ☒ f. flipping horizontally ✓

The correct answers are: adding noise, flipping horizontally, color jittering, dropping out pixels, rotating, zooming and/or cropping

Question 39

Correct

Mark 1.00 out of 1.00

Which of the following statements are true regarding data augmentation?

- ☒ a. Modifications in data augmentation can often be applied on-the-fly. ✓
- ☒ b. Data augmentation is about creating "new" artificial samples by modifying existing samples. ✓
- ☒ c. Modifications in data augmentation heavily depend on the data and task. ✓
- ☐ d. Data augmentation always improves the model performance.

The correct answers are: Data augmentation is about creating "new" artificial samples by modifying existing samples., Modifications in data augmentation can often be applied on-the-fly., Modifications in data augmentation heavily depend on the data and task.

Question 40

Correct

Mark 1.00 out of 1.00

Which of the following statements are correct when talking about normalization/scaling?

- ☒ a. Normalization/scaling can depend on the method/model. ✓
- ☒ b. Normalization/scaling can depend on the data. ✓
- ☐ c. Normalization/scaling only makes sense for image data.
- ☐ d. Normalization/scaling is always beneficial.

The correct answers are: Normalization/scaling can depend on the data., Normalization/scaling can depend on the method/model.

Question 41

Partially correct

Mark 0.75 out of 1.00

Which of the following statements are true regarding loss functions in the area of supervised learning?

- ☒ a. Depending on the task (e.g., regression vs. classification), different loss functions might be appropriate. ✓
- ☒ b. A loss function measures the distance between the model prediction and the true target value. ✓
- ☒ c. Typically, for a given loss function, the lower the loss, the better the prediction. ✓
- ☐ d. Comparing the values of different loss functions (e.g., mean-squared error vs. cross entropy) is usually not meaningful.

The correct answers are: A loss function measures the distance between the model prediction and the true target value., Typically, for a given loss function, the lower the loss, the better the prediction., Depending on the task (e.g., regression vs. classification), different loss functions might be appropriate., Comparing the values of different loss functions (e.g., mean-squared error vs. cross entropy) is usually not meaningful.

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